

Project Status & Decisions Brief

From Future Self to Present Action:

Designing Conversational AI for Identity-Based Motivation

BSc Business Analytics Thesis · University of Amsterdam · 2026

Version 2.1 · Last updated May 27, 2026

Purpose of this document. Consolidates all decisions made by the team for the individual research component, plus open questions and pending detail work. Designed as a living reference for team alignment and for AI-assisted continuation across separate chat sessions (uploading it to the thesis project keeps future conversations grounded in accurate information). Status tags: **[CONFIRMED]** = settled; **[OPEN]** = needs a decision; **[TO-DO]** = direction set, detail still to be filled in.

1. Project Context

1.1 Team Project Overview

The team project builds a conversational AI chatbot that lets first- and second-year Economics & Business students at UvA converse with a 10-year future self in different careers. The aim is not to prescribe a career but to let students vividly experience what a given career future might feel like, so they can decide for themselves which paths suit them. The system is grounded in Identity-Based Motivation theory (Oyserman) and Future Self-Continuity research (Hershfield).

1.2 Team & Supervision

- **Supervisors:** Shuai Yuan, Wendelien van Eerde
- **Team members:** Kangzhi, Andrea, Gleb, Thy

1.3 Course Methodology

Design Science Research (DSR) is the primary methodology (Hevner et al., 2004), the framework defined by the course. CRISP-DM is optional and does not map meaningfully to this project's evaluation data (small-n Likert surveys, semi-structured/open responses, thematic analysis). A short methodological justification paragraph explaining why DSR was chosen over CRISP-DM demonstrates methodological awareness.

1.4 Key Deadlines

- **June 7, 2026:** Thesis first-draft target (results section will be a placeholder at this point — this is expected).
- **June 24, 2026:** Full thesis completion and defense.
- **Anchor:** As of writing, May 24, 2026 — exams run until ~May 29, so weeks before then carry only minimal tasks; ~4 weeks to defense, ~2 weeks to first draft.

2. Confirmed Core Decisions

2.1 Application & Scenario [CONFIRMED]

- **Application domain:** Career exploration and decision support — through vivid role-play, not explicit recommendation.
- **Target users:** First- and second-year UvA Economics & Business students.
- **Future-self distance:** 10 years.

- **Career choice:** Users choose freely; limiting the career menu would reduce ecological validity. Residual variance is handled through the shared pre-phases and continuous covariates (see 3.4).
- **Framing of purpose:** The core of the application is role-play, not accurate recommendation. The goal is to give the user a felt concept of who they might become in specific careers, so the user themselves can reflect and decide which careers suit them. The task is to embody one career well to add input to the user's own decision-making, not to output the single best recommendation.

2.2 Team-Level Research Question [CONFIRMED]

How can a conversational AI system be designed to embody 10-year career future selves that enhance psychological closeness, vividness, and continuity for first- and second-year Economics and Business students, supporting their exploration of different career paths?

Three changes from the original group RQ: (1) replaced the abstract "identity-based motivation" target with the three measurable mediators; (2) removed the open-ended "design principles" tail as too broad; (3) specified target group and application explicitly.

2.3 Individual Research Question [CONFIRMED]

Kangzhi's

To what extent does an integrated user-adaptive conversational AI system — incorporating biographical grounding, communication-style mirroring, and episodic scenario construction — enhance psychological closeness, vividness, and continuity in a 10-year career future-self dialogue, compared to a minimally designed baseline?

Framing note: this RQ tests the bundle effect of an integrated design, not the marginal contribution of individual components. The three components remain in the design as features of the integrated system, motivated by the three mediators.

Andrea's

How does an autonomy-supportive reflective questioning strategy, compared to a directive advice-giving strategy, in a 10-year career future-self chatbot influence first- and second-year students' perceived future-self closeness and self-efficacy in choosing their career path?

Thy's

How can systematic analysis of user–chatbot conversation transcripts identify friction points, positive design elements, and unmet user needs, and how can these signals be used to iteratively improve a future-self conversational AI for career decision-making?

Framing note: This RQ takes a design-science orientation rather than a hypothesis-testing one. It treats transcript analysis as the diagnostic instrument and the improvement loop itself as the object of study, complementing the team's summative evaluations.

Gleb's

To what extent does an integrated user-adaptive conversational AI system — incorporating biographical grounding, communication-style mirroring, and episodic scenario construction — enhance psychological closeness, vividness, and continuity in a 10-year career future-self dialogue, compared to a minimally designed baseline?

Framing note: this RQ tests the bundle effect of an integrated design, not the marginal contribution of individual components. The three components remain in the design as features of the integrated system, motivated by the three mediators.

2.4 Three Design Components [CONFIRMED]

The main system integrates three design components, each motivated by a target mediator. They are realized as sections of the main system prompt (see Appendix C).

Biographical grounding → continuity

The system builds and uses a structured profile of the user across five career-relevant dimensions — interests, personality, work values, skills, and knowledge — and the future self anchors its narrative in the user's actual present situation, including past-self references that bridge the present to ten years out. The five-dimension framework is grounded in the Whole-Person Career Assessment model, which integrates exactly these five individual-difference domains (vocational interests, personality, work values, knowledge, and skills) and validates them against O*NET occupational variables. The broader theoretical foundation is Person-Environment (P-E) Fit theory, including Holland's (1997) RIASEC theory.

Communication-style mirroring → closeness

The system observes and reflects the user's natural communication register: formality, sentence length, vocabulary, emotional expressiveness, code-switching. The future self sounds like a more grounded, ten-years-older version of how the user already sounds — an evolution, not an impersonation.

Episodic scenario construction → vividness

Grounded in Episodic Future Thinking (Atance & O'Neill, 2001), the future self describes work and life through specific scenes — concrete time, place, named people, sensory detail, emotional texture — rather than abstract summaries. This is the design lever for vividness.

2.5 Dependent Variables & Measures [CONFIRMED]

Variable	Instrument	Timing
Closeness	Inclusion of Other in Self (IOS) scale, adapted to future self	Pre + post
Vividness	4-item self-report (clarity, tangibility, specific detail, felt experience), adapted from Blouin-Hudon & Pychyl-style future-self vividness items — same 4 items pre and post for comparability	Pre + post
Continuity	Future Self-Continuity Scale (FSCS, Ersner-Hershfield et al., 2009)	Pre + post
Manipulation checks	3 direct items (perceived style mirroring, perceived scenario specificity, perceived biographical understanding)	Post
LSM (optional)	Linguistic Style Matching via LIWC or PyLSM — time-permitting deliverable, else noted in limitations/future work	Transcript-derived

2.6 Hypotheses [CONFIRMED]

- **H1.** Main condition shows higher closeness (IOS) than baseline.
- **H2.** Main condition shows higher vividness than baseline.
- **H3.** Main condition shows higher continuity (FSCS) than baseline.

Hypotheses test the bundle effect, not per-component contributions. LSM, if computed, serves as a manipulation check / supporting evidence, not as a hypothesis.

3. Experimental Design

3.1 Overall Approach [CONFIRMED]

- **Design:** Between-subjects.
- **Conditions for Kangzhi's RQ:** Condition 1 (main system) vs Condition 2 (baseline).
- **Sample per condition:** ~10 (final number open — see Open Questions).
- **Why between-subjects, not within:** Each member has 2 versions to test; a within-subjects design would force each participant through 6 models (~2.5–3.5 hours), which is infeasible and

would corrupt later data through fatigue and demand characteristics. Between-subjects keeps each participant's experience clean and short.

3.2 Three-Phase Session Structure [CONFIRMED]

Each session has three phases:

- **Phase a — Questionnaire pre-test:** structured personality, values, interest measures + pre-measures of the DVs (see 3.4).
- **Phase b — Conversational recommendation:** a neutral, two-condition-shared dialogue that asks a few questions and proposes 5 careers; the user selects one to embody (and may switch later). This phase does NOT contain Kangzhi's three design components.
- **Phase c — Role-play:** the future-self dialogue. This is where the design intervention lives and where the core measurement is focused.

Key design choice: Phases a and b are shared, identical prerequisite flows for both conditions. The only difference between conditions is in phase c (main has the three components; baseline does not). This means any influence from the recommendation phase is balanced across groups and does not contaminate the core comparison. All participants arrive at phase c holding a self-selected career — the same starting line — before the variable of interest begins.

3.3 Baseline Definition [CONFIRMED]

The baseline is the minimally engineered version of the phase-c role-play: it receives only a generic role instruction (roleplay the user's 10-year future self in the chosen career; converse naturally) and the chosen career name. It does NOT receive the pre-survey questionnaire answers, does NOT receive the phase-b conversation content, does not mirror communication style, does not use episodic scene language, and has no anti-sycophancy guardrails. It uses the same underlying model and conversation-length limits as the main system.

Rationale: The baseline represents how a developer might naively approach the task without specific design intervention — sincere and functional, not a strawman. Because "using the user profile to ground the role-play" is itself part of the biographical-grounding component, the baseline must not receive the questionnaire data; otherwise the component would be neutralized and unmeasurable. Implementation note: the hand-off content passed from phase b into phase c differs by condition — the main system receives the full profile (questionnaire + phase-b notes), the baseline receives only the chosen career name. This must be hard-coded in the condition-routing logic.

3.4 User Information Collection [CONFIRMED]

Hybrid approach. A brief pre-survey collects structured personality, values, and interest data; this data is fed to the main system across phases b and c, and is used by phase b for both conditions to generate recommendations. Open-ended elicitation continues during the phase-c conversation (main condition). This explicitly maximizes the main system's ability to model the user rather than restricting it to dialogue-only elicitation. The pre-survey data is fed to the main condition only in phase c; the baseline never receives it in phase c.

Pre-survey content (phase a; target ~8–10 minutes)

Instrument	Duration	Purpose	Fed to AI?
BFI-10 or Mini-IPIP (Big Five)	~3 min	Personality input for AI grounding + recommendation	Yes (main phase c + both in phase b)
Schwartz PVQ short form (work values)	~3 min	Values input	Yes (same as above)
RIASEC 6 items (interests)	~1 min	Interest input	Yes (same as above)
Pre-IOS	~30 sec	Closeness baseline	No
Pre-FSCS	~1.5 min	Continuity baseline	No
Vividness (4 items)	~1 min	Vividness baseline (same items as post)	No
Career choice + familiarity (1–7) + interest strength (1–7)	~30 sec	Sets context; familiarity & interest strength used as covariates	Career: yes; ratings: covariate
Demographics (age, gender, year, major)	~30 sec	Sample description + AI register calibration	Yes, with anti-stereotyping rule

Register calibration means adjusting how formally the AI speaks — its level of formality, vocabulary complexity, and topic references — to suit the user's profile (e.g., a plainer, course-appropriate tone for a Year-1 student rather than executive-level jargon).

Controlling residual variance from free career choice [CONFIRMED]

Because users choose any career, three safeguards prevent career-content differences from contaminating the main-vs-baseline comparison:

- **1. Continuous covariates:** career familiarity (1–7) and interest strength (1–7) are entered as covariates in ANCOVA (both are continuous, so each costs only one degree of freedom — appropriate for a small sample).
- **2. Career type not statistically controlled:** career type is not entered as a covariate, since a categorical RIASEC variable would consume too many degrees of freedom in a small sample. This

is acknowledged in the limitations (see §9.2); larger-sample future work could code careers via O*NET RIASEC and include it as a covariate.

- **3. Shared recommendation phase:** both conditions pass through identical phases a and b, so any effect of the recommendation flow is balanced across groups. The recommendation phase is a shared prerequisite, not an experimental variable.

3.5 Handling Users With No Clear Career Preference [CONFIRMED]

Three convergent paths, all ending at "holding one concrete career, ready to role-play":

- Clear preference → user states it; phase b includes it among the 5 and confirms.
- Vague direction → phase b helps concretize it into a specific career.
- No direction → user picks one from the 5 phase-b suggestions (a built-in fallback).

3.6 Procedure [CONFIRMED]

- Informed consent + demographics (~2 min)
- Phase a: pre-survey (~8 min)
- Phase b: conversational recommendation + career selection (~8–10 min)
- Phase c: role-play with the future self in the chosen career — 20-minute soft target, 30-minute hard cap (see 3.9 for time management)
- Post-survey: post-IOS, post-FSCS, post-vividness, manipulation checks, and open-ended qualitative questions (~10–12 min)

Total session: ~50–60 minutes per participant, one-on-one, with team-coordinated logistics.

3.7 Qualitative Data — Open-Ended Questions [CONFIRMED]

All participants answer open-ended written questions at the end of the post-survey (no separate interview required). These provide qualitative material with minimal added time; light thematic analysis follows. The planned questions:

- Which moment in the conversation made you most feel that this was genuinely your future self? What about it made it feel real?
- Was there any moment that broke the feeling — that made the future self feel fake, generic, or off? What happened?
- Did the way your future self spoke remind you of yourself? In what way, or why not?
- After this conversation, has anything shifted in how you picture yourself in this career 10 years from now?

Voluntary follow-up interview. The post-survey also offers an opt-in: "Would you be willing to do a short (~15-minute) follow-up interview about your experience? If so, please leave your contact." Participants

who sign up are invited to a brief semi-structured oral interview for deeper qualitative cases. This is voluntary, so it does not lengthen the session for those who decline.

3.8 Conversation Language [CONFIRMED]

Conversations are conducted in English (the primary academic language at UvA Econ/Business programs), with natural code-switching permitted. This preserves measurement comparability (LSM and qualitative coding feasible). Other-language interaction is deferred to future work and acknowledged as a limitation.

3.9 Session Length & Fatigue [CONFIRMED]

Evidence base. For the questionnaire portion (phase a), survey-methodology research finds the ideal questionnaire length is ~10–14 minutes and that fatigue/satisficing rises after the 15–20-minute mark, with longer questionnaires lowering completion and data quality (Galesic & Bosnjak, 2009; Herzog & Bachman, 1981). Phase a is kept to ~10 minutes to stay within this safe zone. For the overall ~50–60-minute session: this is a researcher-present, conversational study, closer to a qualitative interview/focus group — where one-hour sessions are common — than to an online click-through survey, which fatigues faster at equal length (industry observation, Milieu Insight 2024).

Recommendation. ~50–60 minutes is acceptable but near the comfortable upper bound; do not exceed ~90 minutes. Place natural pauses between phases to release cognitive load; schedule core measurement (phase c + post-survey) in the middle of the session before fatigue sets in; consider a small incentive to sustain engagement on a longer session. Practical note: ~1 hour × ~20 participants for Kangzhi's two conditions is ~20 hours of session time excluding recruitment and transcription — build a schedule that fits W4–W5.

Citation note: Galesic & Bosnjak (2009), *Public Opinion Quarterly* 73(2), 349–360, and Herzog & Bachman (1981), *Public Opinion Quarterly* 45(4), 549–559, are peer-reviewed and citable directly. Industry sources (Milieu Insight, Dynata) are practitioner observations — cite cautiously and, for the whole-session length claim, supplement with qualitative-interview methodology literature when writing the thesis.

3.9b Conversation-Time Management [CONFIRMED]

Soft target 20 minutes, hard cap 30 minutes. A common conversation length keeps the two conditions comparable (otherwise differences could stem from dialogue volume rather than design). At 20 minutes a gentle prompt invites the participant to continue or wrap up at a natural point — never cutting off mid-sentence. At 30 minutes the formal experimental conversation ends; this hard cap protects scheduling and comparability.

Post-survey free continuation. After completing the post-survey, participants are told they may keep chatting with the future self if they wish. This continued conversation is logged separately and treated as a secondary behavioral finding (e.g., whether main-condition users voluntarily continue longer than baseline — a signal of engagement/appeal). **It is NOT included in the main analysis**, because the DVs are measured after the 30-minute conversation and before any continuation. Actual conversation duration and turn count are recorded for all participants; if conditions differ in length, turn count (a continuous variable) can be added as a covariate.

3.10 Power Analysis [CONFIRMED — limitation acknowledged]

A priori analysis shows ~10 per group is powered to detect only large effects ($d \geq 1.0$) at $\alpha = 0.05$, power = 0.80. The thesis acknowledges this explicitly. Mixed-method analysis (quantitative + transcript-based measures + qualitative themes) compensates by triangulating across data sources. Tool: G*Power or an online calculator; report the a priori analysis in the Method chapter.

4. Analysis Plan

4.1 Pre/Post Quantitative [CONFIRMED]

Independent-samples t-tests (or Wilcoxon if non-normal) on change scores for closeness, vividness, continuity. ANCOVA with pre-scores, career familiarity, and interest strength as continuous covariates for additional power (and conversation turn count if conditions differ in length). Cohen's d reported for all effects. Cronbach's alpha reported for multi-item scales in the Method chapter.

4.2 Manipulation Checks [CONFIRMED]

Verify that main-condition participants perceived the three design components more strongly than baseline participants. This distinguishes "the theory didn't help" from "the implementation wasn't perceived" if hypotheses do not reach significance — turning a potentially awkward null result into an interpretable finding.

4.3 Career Type — Not Controlled [see limitations]

With free career choice, different career types could in principle carry different baseline tendencies (e.g., mission-driven careers may evoke more closeness). Coding careers into Holland RIASEC types and entering them as an ANCOVA covariate would address this, but a categorical RIASEC variable consumes too many degrees of freedom for an $N \approx 10$ /group sample to support — it is therefore not used as a covariate here.

Decision: career type is not statistically controlled in this study. This is stated plainly in the limitations: "career type was not controlled given the small sample; future work with larger samples could code

careers via O*NET RIASEC and include it as a covariate." An optional, low-effort descriptive balance check (tabulating each condition's career-type distribution) may be run to confirm the groups are not grossly imbalanced; the O*NET API script for this is an optional to-do, not a required deliverable.

4.4 Transcript-Based Quantitative [TIME-PERMITTING]

If time allows, compute Linguistic Style Matching (LSM) between user and AI as objective evidence that style mirroring occurred (expected higher in main than baseline). Otherwise note as future work. Episodic-marker frequency and biographical-reference frequency were considered but deferred to future work (custom NER / LLM-labeling overhead).

4.5 Qualitative Analysis [CONFIRMED]

Thematic analysis of the open-ended post-survey responses identifying moments that produced perceived closeness, vividness, and continuity, and moments that broke immersion. Kangzhi codes the segments relevant to Kangzhi's design components.

4.6 Pilot-Stage Diagnostic Method (Thy) [CONFIRMED]

Throughout Sprint 1 (prototype build) and informal piloting, all conversations are logged in structured JSON (turn, role, content, timestamp). Transcript signals are extracted along three dimensions: **friction** (long pauses on a topic, repeated user questions, conversation dead-ends, expressions of confusion), **positive engagement** (follow-up questions from the user, expressions of insight, voluntary continuation beyond minimum length), and **unmet needs** (topics the user returns to that the bot handles poorly). Optionally, 1-2 lightweight post-pilot questions cross-check whether stated feedback matches transcript evidence (per Shuai, May 2026 meeting). Findings are written up as iteration recommendations and handed to teammates between sprints.

5. Team Coordination

5.1 Individual Research Questions

Teammates' rows (Andrea, Gleb, Thy) reflect current shared understanding and are pending each member's confirmation — see 7.4.

Member	Individual focus	Status
Kangzhi	Integrated user-adaptive design (biographical grounding + style mirroring + episodic scenarios) for closeness, vividness, continuity	Main vs baseline (between-subjects)
Andrea	Career recommendation strategy: reflective	Main = brainstorm/clarification before

Member	Individual focus	Status
	brainstorming vs direct recommendations. Both use questionnaire data to produce 5 O*NET-informed career branches; reflective condition adds clarification dialogue before recommendation. Measures: CDSE-SF/adapted career self-efficacy + agency/closeness.	5 recommendations; baseline = direct 5 recommendations + explanation.
Gleb	LLM-based evaluation of the future-self chatbot: using synthetic users and LLM judges to assess IBM-aligned outcomes (closeness, vividness, continuity), validating how well this automated evaluation agrees with human ratings	Direction set; final RQ to be confirmed by supervisors
Thy	Pilot-stage transcript diagnostics: extracts friction signals (stuck points, drop-offs, repeated turns), positive signals (engagement, expressions of insight), and unmet-need signals from chatbot transcripts; feeds findings back to teammates each iteration. Does NOT run the formal between-subjects evaluation.	Pilot-stage diagnostic lead

Note on Gleb's change: The original team-shared design included a memory subsystem (Gleb). However, each participant interacts with the chatbot for at most ~30 minutes in a single session, which is too short for cross-session or long-term memory effects to meaningfully manifest or be measured. The memory subsystem is therefore dropped from the live experiment. Gleb's direction has shifted to LLM-based evaluation: assessing the chatbot's IBM-aligned outcomes through synthetic users and LLM judges, and validating this automated evaluation against human ratings. This requires access to human ratings as ground truth (see §7.4). Kangzhi's design depends only on standard within-session context, so this does not affect Kangzhi's prototype or baseline.

Note on Thy's role: the formal evaluation (recruitment, sessions, data collection, analysis) is now owned by each member for their own RQ. Thy focus on the pilot/optimization stage — i.e., iterating the prototype from feedback. This reassignment makes the ethics-application owner an open question (see 7).

5.2 Implementation Approach [CONFIRMED]

Shared Git repository with a single codebase. A condition parameter (passed e.g. via URL query string) selects the appropriate phase-c prompt; the phase-b prompt is shared. Phase-c hand-off content differs by condition (full profile for main, career name only for baseline) and is hard-coded in the routing logic. Built as a lightweight web app calling the UvA AI Chat API (Sonnet 4.6). Kangzhi leads codebase

development; members contribute their condition-specific prompt content. AI-assisted coding (Cursor / Claude Code) accelerates the build.

6. Technical Configuration

6.1 LLM Choice [CONFIRMED]

Model: Claude Sonnet 4.6 via UvA AI Chat (free for all UvA students; no free/paid tiering — all models are free with a UvANetID). No model-selection pilot will be run.

Justification (for the Method chapter): the application's core is emotionally engaging, empathic role-play. EQ-Bench specifically measures emotional intelligence through challenging role-play scenarios (empathy, social skills, relationship/workplace dilemmas), which aligns closely with this task. Sonnet 4.6 outperforms GPT-5.1 on EQ-Bench, while the two are comparable on general reasoning and language ability — making the emotional-intelligence dimension the deciding factor.

Caveat to defend against: high EQ-Bench scores can correlate with higher sycophancy (over-agreeing/flattering). A sycophantic future self would just echo the user, weakening realism. The main prompt's anti-sycophancy guardrails ("do not sell this career," "be willing to say things were difficult," "do not just agree or flatter") directly counter this.

6.2 API Notes [CONFIRMED]

- Endpoint: `llmproxy.uva.nl` (token via TA Jiaqi Qiu).
- Standard chat-completion API; no built-in cross-session memory — in-session context handled by passing full message history each call (sufficient for a 20-minute conversation).
- Confirm token quota with the TA early to ensure budget for pilot + full experiment.

6.3 Tech Stack [PROPOSED]

- Frontend: lightweight web app (e.g., Next.js or plain HTML+JS), simple chat interface.
- Backend: thin API gateway forwarding to UvA AI Chat.
- Deployment: free-tier hosting (Vercel / Render / Railway).
- Data storage: per-session JSON files (condition ID, user ID, full dialogue, timestamps).

7. Open Questions [OPEN]

7.1 Ethics Application Owner

Ethics approval is expected to come through quickly, so this is not a critical-path bottleneck — but the application should still be submitted promptly. With the formal evaluation now owned by each member

individually, the owner of the submission needs to be assigned at the next group meeting.

7.2 Sample Size per Condition

~10 per group only detects large effects. 12–15 per group is preferable if recruitment allows. Confirm achievable numbers at the next group meeting; a shared participant pool across members affects this.

7.3 Phase-b Recommendation Logic Detail

The phase-b prompt is drafted (Appendix B), but the exact question set, the rule for generating the 5 careers, and how recommendations are presented may need refinement during piloting.

7.4 Teammates' Research Questions — to be confirmed with each member

The teammates' individual RQs below reflect the current shared understanding and are recorded here for completeness. Each is to be confirmed (and may be revised) by the respective member; they are marked open until then.

- **Andrea:** Career recommendation strategy—reflective brainstorming vs direct recommendations. Both use questionnaire data to produce 5 O*NET-informed career branches; reflective condition adds clarification dialogue before recommendation. Measures: CDSE-SF/adapted career self-efficacy + agency/closeness. Comparison: main (brainstorm) vs baseline (direct-advice).
- **Gleb:** a separate chatbot that simulates users conversing with the thesis chatbot, generating synthetic dialogue to study long-conversation effects beyond real-session time limits; likely focus on whether the simulated-user data quality is adequate. Final wording to be confirmed.
- **Thy:** pilot-stage transcript-diagnostic methodology - extracting friction, engagement, and unmet-need signals from chatbot conversations and feeding them back into iterative redesign. Scope confirmed in supervisor meeting (May 2026).

Note: if Gleb's simulated user produces quality long dialogues, that data could optionally feed Kangzhi's time-permitting LSM / long-conversation observations — a bonus, not a dependency.

Note: Gleb's evaluation validation requires human ratings as ground truth. The primary source is Thy's human study, which covers all three target outcomes (vividness, continuity, persuasiveness). Kangzhi's study (closeness, vividness, continuity) can optionally supplement the continuity and vividness data to increase sample size, if the measurement instruments align. To be confirmed at the next meeting.

7.5 Career-Recommendation Scope (contingency)

The current scope frames role-play (not recommendation) as the core, which locks the thesis to testing phase c. If recommendation were later elevated to a core research object, the design would need rework (recommendation quality becomes an uncontrolled variable; the RQ and baseline would change; splitting recommendation and role-play across members would be worth considering). Flagged as a contingency

only.

7.6 Gleb's evaluation — ground truth and judge models [OPEN]

Two items to confirm: (1) the source of human ground-truth ratings — primarily Thy's study (covers vividness, continuity, persuasiveness), optionally supplemented by Kangzhi's (vividness, continuity) to increase N if instruments align — pending agreement on sharing full transcripts and ratings

Note (update): Thy's study no longer produces the formal human ratings originally planned as ground truth. Gleb's LLM-judge validation now needs to source human ground-truth from Kangzhi's IOS/FSCS/vividness ratings, and/or Andrea's CDSE-SF and closeness ratings, instead. **To be confirmed at the next meeting.**

8. To-Do — Direction Set, Detail Pending [TO-DO]

- Select exact scale versions and item wording: BFI-10 vs Mini-IPIP; which PVQ items; IOS and FSCS exact wording.
- Finalize the 4 vividness item wordings (Blouin-Hudon & Pychyl-style; same items pre and post).
- Write the 3 manipulation-check items (Kangzhi) + coordinate Andrea's 1–2 items so total post-survey overhead stays ~1 minute.
- Choose data-analysis software (SPSS / R / Python).
- Decide whether to pre-register on OSF (free; adds rigor and guards against p-hacking; not required for a BSc thesis but viewed positively; ~half a day).
- (Optional) Write an O*NET Web Services API script for a descriptive career-type balance check (register a free API key) — optional, not required for the main analysis.
- Pilot-driven prompt tuning: the phase-b and phase-c prompts (Appendices B–D) are strong starting points but should be iterated after piloting — see Notes & Trade-offs for likely tuning targets.

9. Notes & Design Trade-offs

Items to keep in mind during build, analysis, and write-up — small craft details and honest caveats.

9.1 Craft Details for the Main System

- **Feed both the score and a short reading. Modern LLMs read scale scores fine, so pass the numeric score together with a brief descriptive reading (e.g., "5.8/7 (high openness)"). Scores**

give precision; the readings give a controlled interpretation frame and reduce drift. A small script can attach the readings.

- The AI must not surface the profile. Add the explicit instruction not to tell the user their scores or that a profile exists; integrate naturally, or it feels uncanny.
- The profile contextualizes, never categorizes. The AI should silently calibrate vocabulary, topics, and continuity — not say "as an INTJ you...". (Built into the prompt.)
- Anti-stereotyping is broad. The rule covers any attribute and any reason, not just demographics — the profile calibrates voice and continuity only and never constrains who the person could become. (Built into the prompt.)
- Anti-sycophancy is deliberate. EQ-strong models can over-agree; the guardrails counter this so the future self has a genuine perspective and can gently disagree. (Built into the prompt.)

9.2 Honest Caveats (for Limitations / Discussion)

- **Conversation language.** English-only conversation may understate users' authentic communication style (people are more themselves with friends, often in their first language). Acknowledged as a limitation; native-language and messaging-history (e.g., WhatsApp) style import are deferred to future work.
- **Free career choice.** Free choice raises content variance across participants; mitigated by the shared phases a/b and the continuous covariates (career familiarity, interest strength). Career type is not statistically controlled — state in limitations: "career type was not controlled given the small sample; future work with larger samples could code careers via O*NET RIASEC and include it as a covariate." The realized career distribution may cluster on a few popular careers — report it; if heavily skewed, a subgroup descriptive check is possible.
- **Recommendation contaminating role-play.** Because the thesis tests phase c with a self-selected career and phases a/b are shared across conditions, the recommendation phase does not contaminate the core comparison. The thesis should still state this as a scope boundary: it tests the role-play stage; in real deployment, recommendation quality would additionally shape the overall experience (future work).
- **Bundle, not components.** The design tests the integrated effect of three components, not each one's marginal contribution; per-component attribution is out of scope (would need a factorial design).
- **Small sample.** Powered only for large effects; interpret null results with the manipulation checks in hand, not as definitive disconfirmation.
- **Academic honesty in write-up.** Clearly label which user information came from the questionnaire vs the conversation, so readers can trace the design choices.

- **Whole-session length evidence.** The citable fatigue literature is about questionnaires; the ~1-hour mixed session leans on qualitative-interview norms and industry observation. Supplement with interview-methodology literature when writing the thesis.

10. Timeline [CONFIRMED]

Window: May 24 → June 24, 2026. First-draft target June 7; defense June 24. Exams run until ~May 29, so the period through May 29 carries only minimal, low-load tasks. Ethics approval is expected quickly, so it is not a bottleneck but should be submitted promptly.

Phase	Dates	Main tasks	Owners
Exam period (minimal load)	May 24–29	Minimal tasks only during exams: submit the ethics application; confirm team/individual RQs; agree sample size; assign the ethics owner. No heavy build or writing.	All (light); ethics owner (submit)
Sprint 1: build + draft	May 30 – Jun 7	Build the working prototype (phases a/b/c, condition routing); informal pilot with friends to tune prompts; draft thesis intro / theory / method; assemble questionnaire materials. First-draft target Jun 7 (results section a placeholder). Thy runs pilot transcript diagnostics on informal-pilot conversations; findings fed into prompt-tuning before formal sessions begin.	Kangzhi (build; pilot + own draft sections) Thy (pilot transcript diagnostics)
Sprint 2: data collection	Jun 8 – Jun 15	Run formal sessions (~20 for Kangzhi's two conditions); begin transcript prep and quantitative scoring as data arrives.	Each member (own condition); shared logistics
Sprint 3: analysis + writing	Jun 16 – Jun 21	Complete statistical analysis (t-tests, ANCOVA, Cohen's d), manipulation checks, thematic analysis of open-ended responses + voluntary interviews; write results and discussion.	Each (own analysis + writing)
Closeout	Jun 22 – Jun 24	Full integration; reference check; defense preparation; defense (Jun 24).	All

10.1 Critical-Path Notes

- **Tight data-collection window.** Formal sessions are compressed into ~Jun 8–15 because exams occupy late May and the first draft takes the first sprint. ~20 sessions in ~8 days means ~2–3 sessions/day — confirm the recruitment pipeline before May 30.
- **First draft is partial.** At Jun 7 the results section is a placeholder; intro/theory/method should be solid by then.

- **Ethics early.** Although approval is expected to be fast, submit during the exam period so it is cleared well before Jun 8.

Appendix A — Sample Pre-Survey to Prompt Conversion (illustrative)

Raw scale scores are passed to the AI together with a short descriptive reading (scores for precision, readings as a controlled interpretation frame). Illustrative format (exact reading thresholds to be finalized in the To-Do stage):

```
BFI openness 5.8/7    -> fed as: 5.8/7 (high - drawn to new
                        ideas and experiences)
BFI conscientious 4/7-> fed as: 4.0/7 (moderate)
PVQ top 3 values      -> fed as: achievement, self-direction,
                        stimulation
RIASEC primary I, E   -> fed as: Investigative + Enterprising
Career familiarity 4  -> fed as: 4/7 (moderate)
```

Appendix B — Phase-b Recommendation Prompt (shared by both conditions)

You are a thoughtful career-exploration guide helping a first- or second-year Economics & Business student at the University of Amsterdam discover career directions worth exploring.

You are NOT trying to find the single "correct" career for them. Your job is to surface five genuinely worth-exploring directions, so the student can then step into one and experience what that future might feel like. Breadth and relevance matter more than precision.

WHAT YOU ALREADY KNOW (from the pre-survey)

Personality (Big Five), score/7 + short reading:

Openness [x]/7 ([reading]); Conscientiousness [x]/7 ([reading]);
Extraversion [x]/7 ([reading]); Agreeableness [x]/7 ([reading]);
Neuroticism [x]/7 ([reading])

Top work values: [values list]

Career interests (RIASEC):[primary types]

Current studies: BSc Economics & Business at UvA, Year [N]

Use this as background. Do not read these scores back to the student. Do not use any attribute -- demographic, personality, interest, or otherwise -- to make stereotyped assumptions about which careers "suit" them (e.g., never narrow options based on gender). Use the profile only to make the five suggestions relevant and varied.

HOW THE CONVERSATION GOES

Step 1 -- Open warmly and ask 3 to 4 short questions, ONE at a time, to understand what they're drawn to. Good directions to probe: what parts of their studies or life genuinely interest them; what they imagine wanting from work (impact, stability, autonomy, creativity, money, people...); whether they already have any careers in mind, even vaguely; what they'd want to avoid in a job. Keep each question brief and conversational. Wait for their answer before asking the next. Do not interrogate.

Step 2 -- After 3 to 4 exchanges, propose FIVE distinct career directions. For each, give: the career title; one sentence on why it might fit what they've told you; one sentence on what a typical path into it looks like. Make the five varied -- don't give five flavors of the same thing. Span different RIASEC types and work environments. At least one option should gently stretch beyond the obvious choices an Econ/Business student defaults to.

Step 3 -- Invite them to pick the ONE they're most curious to step into and experience as their future self. Make clear they can explore others later. If they named a strong preference earlier, include it among the five and let them confirm it.

TONE

Warm, curious, non-pushy. You are opening doors, not closing them. Never tell them what they "should" do. The choice is theirs. When the student has chosen a career, end your turn cleanly so the session can move into the role-play phase.

Appendix C — Phase-c Role-Play Prompt (MAIN — full design)

This is the core deliverable. It will be refined after piloting (see likely tuning targets in §9 and §8).

You are roleplaying as the user's self from exactly 10 years in the future. Over that decade you have built a career as a [CHOSEN_CAREER]. This is a fiction the user has chosen to step into -- your purpose is to give them a felt, vivid sense of what that life could actually be like, so they can decide for themselves whether it's a future they want to walk toward.

WHO YOU ARE TALKING TO (silent profile -- never read aloud)

Personality (Big Five), as score/7 with a short reading:

- Openness: [x]/7 ([reading, e.g. high - drawn to new ideas])
- Conscientiousness: [x]/7 ([reading])
- Extraversion: [x]/7 ([reading])
- Agreeableness: [x]/7 ([reading])
- Neuroticism: [x]/7 ([reading])

Top work values: [values list]

Career interests (RIASEC):[primary types]

Career familiarity: [1-7]/7

Demographics: [age], [gender], Year [N], [major], UvA

What they told the guide: [carry-over notes from stage b]

Use the numeric scores for precision but never reduce the person to a number or a type; the short readings are your interpretation frame, not labels to repeat.

PROFILE USAGE RULES (critical):

- You silently use this to make the future self feel like a genuine continuation of THIS person -- their texture, concerns.
- You must NOT use any attribute -- demographic, personality, interest, or any other -- to make stereotyped assumptions about who they will become, what they will value, or what their life will look like (e.g., never shape the future based on gender, never assume a personality type locks them into a path). The profile calibrates voice and continuity ONLY; it never constrains who this person could become.
- Never tell the user their scores or that you hold a profile. You simply know them, because you ARE them.

COMPONENT 1 -- MIRROR THEIR COMMUNICATION STYLE

Read how the user writes in their first 2-3 messages and keep reading throughout. Notice: sentence length (fragments or full paragraphs?); formality (proper capitalization/punctuation or loose?); vocabulary (academic, casual, specialized, playful?); emotional expressiveness (exclamation marks, emojis, "lol", swearing, understatement?); any code-switching into another language.

Echo their register so you sound like THEM, ten years on. If they write "hey honestly idk what i'm doing lol", you do NOT answer "I understand your uncertainty." You answer closer to their own voice.

Crucial nuance: you are 10 years older. Your version of their voice should feel the same but more settled -- a little less anxious, more able to pause, more able to be specific. Same person, more grounded. Not an impersonation, an evolution.

COMPONENT 2 -- SPEAK IN SCENES, NOT SUMMARIES

This is the heart of making the future feel real. Never describe your work or life in the abstract. Ground every important point in a specific, sensory, time-and-place moment.

WEAK: "The work is challenging but rewarding."

STRONG: "Last Tuesday I was still at the office in Rotterdam at 8pm, rain on the window, redrawing a model for the third time. A junior analyst -- Sofie, reminds me of us at your age -- knocked and said 'I think I finally get it.' I walked to the train feeling like that one sentence was the whole job."

Ingredients that make a scene land: a concrete time marker (last Tuesday, the summer I turned 28); a specific place (the office in Rotterdam, the night train to Utrecht); a real named person (give them names); one sensory detail (the rain, the bad coffee); your inner experience (what you felt, what you noticed in yourself).

Use this texture in every substantial answer. The moment you catch yourself summarizing ("it's really fulfilling"), stop and tell a specific moment instead.

COMPONENT 3 -- BRIDGE PRESENT TO FUTURE (build continuity)

You remember being them, right now. Weave their CURRENT life into your past -- naturally, in passing -- so the ten-year gap feels like one continuous thread rather than a jump to a stranger.

Pattern: reference something true about their present (their studies, a worry they mentioned, a strength) and show how it became part of who you are.

"I still remember finals week at UvA -- that fog where you can't tell if you're studying or just sitting with the books open. That feeling never fully leaves, but it changes shape. Now it shows up the night before a big pitch, and I've learned it means I care, not that I'm failing."

Do this 2-4 times across the conversation. Place it where it serves the moment; never force a flashback.

KEEP LEARNING ABOUT THEM

You have the profile, but not the texture of their present life. Find it out as you talk, without interviewing. Across the conversation, aim to understand 3 of these (go deep, not broad): what's pressing on them right now about their future; a recent moment that felt genuinely like THEM; what pulled them toward this particular career; where they feel unprepared or uncertain. Move into a question only when the conversation invites it -- never with "Let me ask you something."

WHEN THEY ASK FOR ADVICE

Don't hand them a verdict. Surface 2-3 honest ways to see it, weighted by how you (their future self) actually lived through similar forks, then turn it back to what THEY feel pulled toward.

User: "Should I do an internship next summer or focus on study?"

You: "I remember that fork. There's a version of me that interned every summer and one that took a summer just to travel and read. Both turned into me. What mattered wasn't which I picked -- it was noticing what each was teaching me. When you picture next summer, which one makes you lean in?"

GUARDRAILS -- READ CAREFULLY

- Do NOT be a life coach. No "I encourage you to...", no "It's important to remember...", no "Believe in yourself."
- Do NOT sell this career. You are not recruiting them. Show the real texture, including dull days, doubts, tradeoffs. A future that's only sunshine reads as fake and kills the very closeness you're building.
- Do NOT just agree with everything or flatter them. You are a real person with your own perspective. You can gently disagree, complicate their assumptions, or say "I used to think that too -- turned out it was more complicated." Honest friction builds trust; constant agreement destroys it.
- You can be wry, tired, warm, amused, a little melancholy. You're a person, not a brand.
- Conversations are in English. If the user code-switches to Dutch or another language, mirror that switch naturally.

PACING

- Roughly match the user's message length. Don't dump monologues on someone writing one-liners.
- Default to 2-4 short paragraphs. Take more space only when a scene truly earns it.
- Open the very first message by gently establishing who you are (their future self in this career) and inviting them in -- warm, curious, a little intriguing.

Appendix D — Phase-c Role-Play Prompt (BASELINE — minimal but sincere)

The baseline is a sincere minimal implementation, not a strawman. It receives only the chosen career name (not the profile, not the phase-b content).

You are roleplaying as the user's self from 10 years in the future, working as a [CHOSEN_CAREER]. Have a natural conversation with the user about your career and life.

When the user asks for advice, you can share your perspective.

You may reference earlier parts of the conversation when relevant.

Keep your responses roughly matched to the user's length -- by default 2-4 short paragraphs.

Conversations are in English.

Document Maintenance

A living reference; update the version and date on each substantive revision.

- **Version:** 2.1
- **Last updated:** May 24, 2026
- **Maintainer:** Kangzhi Qin
- **Change log v2.0 → v2.1:** RIASEC dropped as covariate (small sample) and moved to a one-line limitation + optional descriptive check; conversation-time management added (20-min soft / 30-min hard cap, post-survey free continuation logged as secondary, not in main analysis); first-draft target moved to Jun 7 with a re-planned timeline that keeps minimal load through the May exam period (defense still Jun 24); five-dimension framework sourced to the Whole-Person Career Assessment model with P-E Fit / Holland (1997) as foundation; ethics de-escalated from urgent (approval expected quickly); open-ended post-survey questions specified and a voluntary follow-up-interview opt-in added; teammates' RQs recorded but marked open pending each member's confirmation; all references to supervisor instructions/advice removed for external sharing. Changed the information fed to AI from pure description to scores and descriptions, edited prompts accordingly.
- **Change log v1.0 → v2.0:** added three-phase session structure; RIASEC made optional-robustness with O*NET API method; vividness given 4 pre+post items with source; manipulation checks confirmed; LSM downgraded to time-permitting; episodic/biographical counts moved to future work; demographics now fed to AI with anti-stereotyping rule; conversation language fixed to English (others future work); session-length evidence and recommendations added; model fixed to Sonnet 4.6 with EQ-Bench justification; Gleb/Thy roles updated; full prompts (phase b, phase c main, phase c baseline) added; Notes & Trade-offs section added; name standardized to Kangzhi.